WSABI Summer Research Lab **Summer 2026**

Lecture 9: The Bootstrap

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9.1 Example: Expected Points in Football

Last week, we modeled **expected points (EP)** in football by predicting the next scoring event from game state variables such as yardline, down, distance, and time remaining, then converting those event probabilities into an expected-points value.

Let S_i denote the game state on play i, and let Y_i denote the next scoring event. If the possible next scoring events are encoded by point values v_c, then the fitted expected points at state s are

$$\widehat{EP}(s) = \sum_c v_c \widehat{\mathbb{P}}(Y = c \mid S = s).$$

Start with a concrete question:

> What is a 95% confidence interval for expected points at `1st-and-10` on the opponent 35?

Or, more generally:

> How uncertain is a fitted EP curve, or the gap between two points on that curve?

Write the quantity of interest as a statistic $T(D)$ computed from the observed dataset D. In different problems, $T(D)$ might be a sample mean, a sample median, a 90th percentile, a regression slope, a predicted EP value, or a contrast between two fitted values.

Here the target is a nonlinear function of many fitted probabilities, so there is no short clean confidence-interval formula to derive by hand.

9.2 What We Wish We Could Do

Ideally, we would repeatedly observe many parallel football seasons generated by the same underlying process, refit the EP model each time, and look at how the fitted quantity of interest moves around.

For example, let

$$T(D) = \widehat{\theta} = \widehat{EP}(s_0),$$

where s_0 is a fixed game state such as `1st-and-10 at the opponent 35`. If we could repeatedly redraw datasets

$$D^{(1)}, \ldots, D^{(B)}$$

from the true data-generating process, we could compute

$$T(D^{(1)}), \ldots, T(D^{(B)}),$$

and use their spread to understand the sampling variability of $T(D)$.

The problem is that we only have one observed dataset. The bootstrap replaces the impossible repeated-sampling experiment with a feasible one: treat the observed dataset D as an empirical stand-in for the population, repeatedly resample from it, and recompute the statistic.

This gives bootstrap datasets

$$D^{*(1)}, \ldots, D^{*(B)}$$

and corresponding bootstrap statistics

$$T^{*(1)}, \ldots, T^{*(B)}.$$

We then use the variation across the bootstrap statistics as an approximation to the variation we would have seen across truly new datasets.

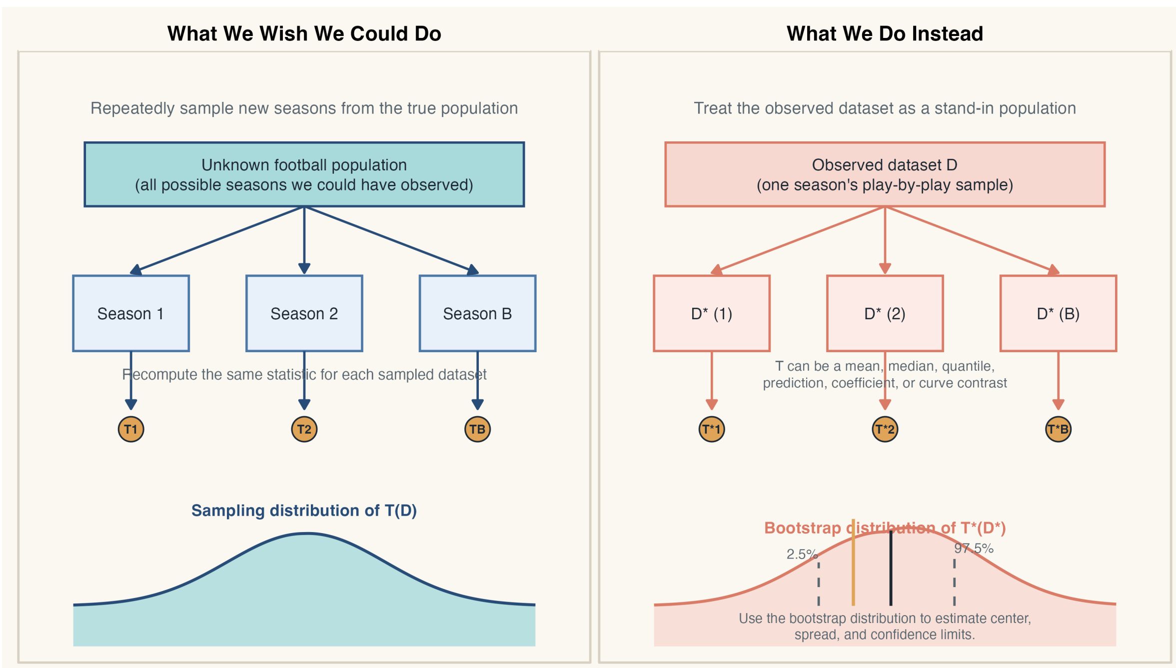

Figure 9.1: Left: repeated sampling from the true population gives the sampling distribution we want. Right: bootstrap resampling from the observed dataset gives a bootstrap distribution that we use as an approximation.

9.3 A First Bootstrap for Expected Points

Now write the observed dataset as D, and let $D^{*(b)}$ denote bootstrap sample b. Then the bootstrap statistic is

$$T^{*(b)} = T\left(D^{*(b)}\right).$$

For expected points, $T(D)$ might be $\widehat{EP}(s_0)$, but it could just as easily be a fitted difference

$$\widehat{\Delta} = \widehat{EP}(s_{20}) - \widehat{EP}(s_{40}),$$

or a sample mean, sample median, or upper quantile.

Suppose our EP dataset has n observed plays. A basic **observation bootstrap** proceeds as follows:

1. Sample n plays with replacement from the observed play-level dataset.
2. Refit the EP model on that bootstrap sample.
3. Recompute the target quantity, such as
$$T^{*(b)} = \widehat{EP}^{*(b)}(s_0).$$
4. Repeat for $b = 1, \ldots, B$ bootstrap replications.

This gives a bootstrap distribution
$$T^{*(1)}, T^{*(2)}, \ldots, T^{*(B)}.$$

The same mechanics work whether T is a mean, median, quantile, prediction, coefficient, or curve contrast. Once we have the bootstrap distribution, we can use it in several ways:

- The bootstrap mean or median gives a picture of the typical re-estimate under resampling.
- The bootstrap standard error measures spread:
$$\widehat{SE}_{boot}(T) = SD\left(T^{*(1)}, \ldots, T^{*(B)}\right).$$
- The bootstrap quantiles give interval estimates, such as the percentile interval
$$[T^*_{0.025}, T^*_{0.975}].$$
- More generally, any bootstrap quantiles can describe skewness, tail behavior, or other features of uncertainty that one standard error would miss.

Usually the original statistic $T(D)$ remains our point estimate, but the bootstrap distribution tells us how much it tends to move around and whether that movement is symmetric or skewed.

9.4 Why the Resampling Unit Matters

The play-level bootstrap above is the easiest version to explain, but it sneaks in a strong assumption: it treats plays as if they were i.i.d.

That is not quite right in football. Plays are grouped within drives, games, teams, and seasons. Nearby observations share context, personnel, score effects, and strategy. So a naive row bootstrap may understate uncertainty by pretending the data are more independent than they really are.

This is the main lesson for the bootstrap:

> Do not just resample rows mechanically. Resample the unit that best matches the dependence structure of the data.

For expected-points data, more defensible choices might be:

- resample whole **drives**,
- resample whole **games**,
- or, in a more time-series setting, resample **contiguous blocks** of observations.

The EP example already shows why different bootstrap variants exist.

9.5 Which Bootstrap Are We Actually Using?

The bootstrap is not one single procedure. It is a family of related procedures, chosen to match the structure of the problem. Each version keeps the same skeleton from Figure 9.1: create a bootstrap dataset, recompute the same statistic T^*, and summarize its distribution. What changes is the resampling scheme.

9.5.1 Observation Bootstrap

This is the default bootstrap: resample observations or rows with replacement. It is the right first choice when the observational units are plausibly i.i.d.

The key assumption is that the rows themselves are exchangeable draws from one common population. In the simplest version, that means i.i.d. observations.

If $T(D)$ is a sample mean, sample median, 90th percentile, or fitted EP value, we simply recompute that same statistic on each resampled dataset. For expected points, observation bootstrap means resampling individual plays.

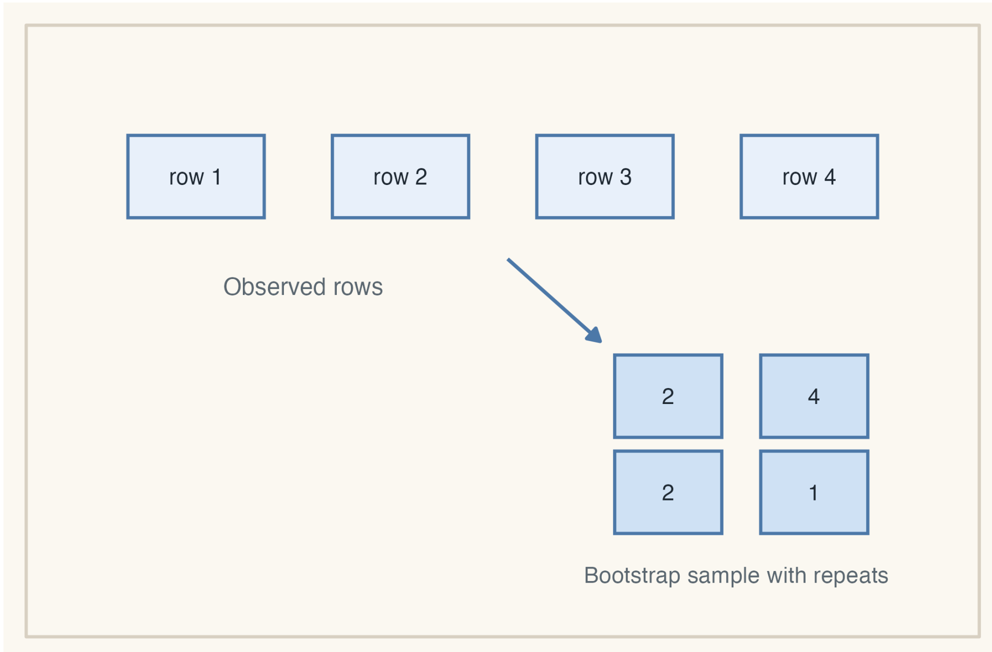

Figure 9.2: Observation bootstrap resamples rows with replacement, then recomputes the same statistic on each resample.

9.5.2 Cluster or Block Bootstrap

If dependence lives inside groups or stretches of time, naive row resampling is too optimistic. In play-by-play football data, resampling whole **drives** or **games** is often more sensible than resampling single plays. In a genuine time series, one often resamples **contiguous blocks** so that short-run dependence is preserved.

The key assumption is no longer row-wise independence. Instead, dependence is allowed within a cluster or within a short block, while the resampled clusters or blocks are treated as the units that can be shuffled around.

Conceptually, cluster and block bootstrap are doing the same thing: preserve the dependence structure you believe is real, then recompute the same mean, median, quantile, prediction, or contrast on each resample.

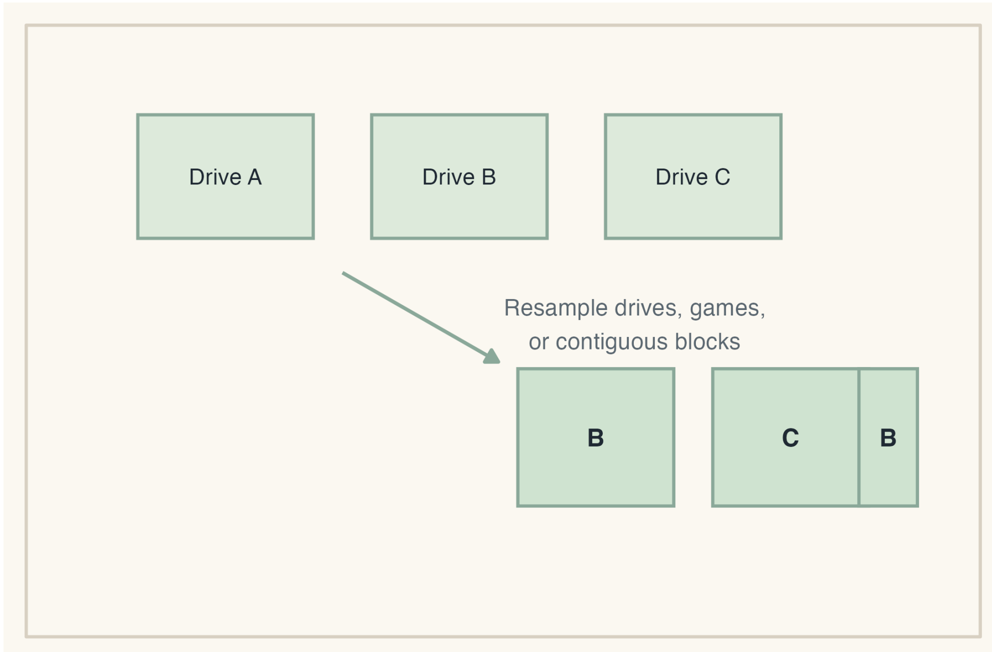

Figure 9.3: Cluster or block bootstrap changes the resampling unit so that dependence is carried into the bootstrap samples.

9.5.3 Parametric Bootstrap

Suppose we trust the fitted EP model itself. Then we can simulate from that model rather than resampling observed rows.

The key assumption here is model correctness, or at least that the fitted model is a reasonable approximation to the data-generating mechanism.

For expected points, this would look like:

1. Fit the multinomial next-scoring model once.
2. For each observed state S_i, simulate a new next scoring event Y_i^* from the fitted conditional probabilities.
3. Refit the EP model to the simulated dataset.
4. Recompute the target $\widehat{\theta}^{*(b)}$.

This is a **parametric bootstrap**. It is more model-dependent than the observation bootstrap, but often very natural when the fitted model is taken seriously.

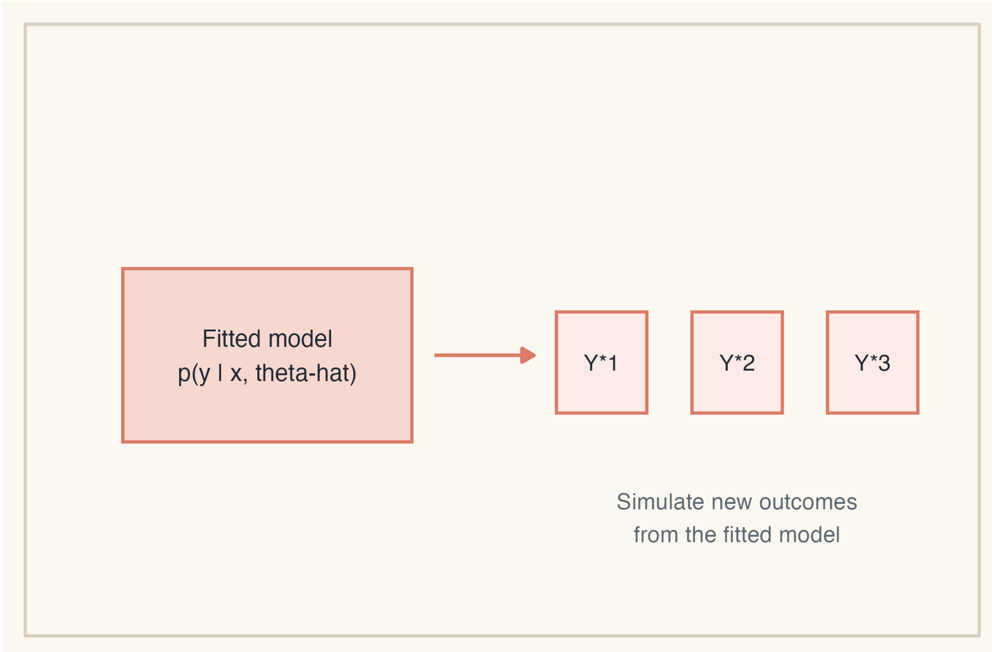

Figure 9.4: Parametric bootstrap simulates new outcomes from the fitted model, then refits and recomputes the statistic.

9.5.4 Residual Bootstrap

Residual bootstrap is most natural in fixed-X regression settings, especially linear models.

The key assumptions are that the design points X_i are treated as fixed and that the regression errors are roughly i.i.d. around the fitted mean. With raw residual resampling, constant variance is usually part of that story as well.

- Fit the model once and compute fitted values $\widehat{Y}_i$ and residuals e_i.
- Resample residuals with replacement.
- Form new responses $Y_i^* = \widehat{Y}_i + e_i^*$.
- Refit and recompute the statistic of interest.

This is less natural for expected points, but useful in regression settings where uncertainty is measured conditional on the observed design matrix. If heteroskedasticity is a serious concern, plain residual bootstrap can be a poor choice. The final summaries are the same: we still look at the distribution of the recomputed statistic and report its mean, median, standard error, or quantiles as needed.

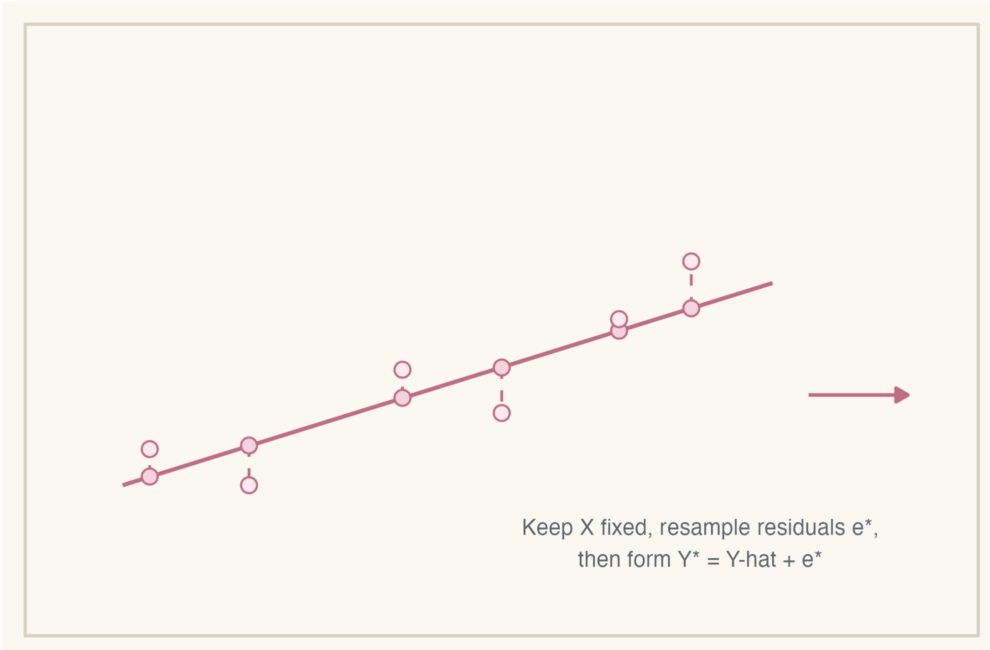

Figure 9.5: Residual bootstrap keeps the design points fixed, resamples residuals, and builds new responses before refitting.

9.6 When the Bootstrap Helps Most

The bootstrap is especially useful when:

- there is no short, reliable formula for the standard error or confidence interval you want,
- the target is a summary like a median, an upper quantile, or some other statistic that is not as convenient as a sample mean,
- the target is built from many fitted pieces, such as a nonlinear function of several estimated probabilities,
- the quantity of interest is a prediction, a contrast, or a feature of a fitted curve rather than one simple coefficient,
- or you can recompute the statistic easily on a resampled dataset even though its sampling distribution is hard to derive analytically.

Here the object of interest is not just one coefficient from a simple formula. It is a fitted functional object built from many class probabilities.

9.7 Limits

The bootstrap is powerful, but it is not automatic.

- If the resampling unit is wrong, the uncertainty estimate can be wrong.
- If the sample is small or unrepresentative, the empirical distribution may be a poor proxy for the population.
- If the statistic is very nonsmooth, such as an extreme order statistic (like the maximum or min), the basic bootstrap can perform poorly.
- The bootstrap does not fix confounding, bad targets, or bad model specification.

In short, the bootstrap approximates sampling variability by resampling and recomputing, but the answer is only as good as the resampling scheme.

Takeaways

- The bootstrap replaces the unknown population with the observed dataset and approximates sampling variability by repeated resampling and refitting.
- The object being bootstrapped is a statistic $T(D)$, which might be a mean, median, quantile, prediction, coefficient, or fitted EP contrast.
- For expected points, a confidence interval for a fitted EP value, contrast, or curve is not easy to write down analytically.
- A naive play-level bootstrap is easy to describe, but sports data often call for a drive-level, game-level, or block-style resampling scheme instead.
- Parametric bootstrap simulates from the fitted model; residual bootstrap is most natural for fixed-X regression problems.
- The resampling unit should mimic the dependence structure you believe generated the data, and the bootstrap distribution can then be summarized with means, medians, standard errors, or quantiles.

References

[BYW] Brill, R. S., Yurko, R., & Wyner, A. J., *Analytics, Have Some Humility: A Statistical View of Fourth-Down Decision Making*, The American Statistician, 2025.

[ET] Efron, B., & Tibshirani, R. J., *An Introduction to the Bootstrap*, Chapman and Hall, 1993.